H2 ANDERSON JUNIOR COLLEGE
HIGHER 2

CANDIDATE NAME

PDG | | | | | PDG INDEX NUMBER | | |

I

BIOLOGY 9648/03

Applications Paper and Planning Question
Paper 3

14 September 2015
Monday

2 hours

Additional Materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your name and PD group on all the work you hand in.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graph or rough working.
Do not use paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

All working for numerical answers must be shown.
At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

Calculators may be used

For Examiner's Use	
1	
2	
3	
4	
5	
Total	**72**

This document consists of **16** printed pages.

Answer **all** the questions.

1 In 2008, a woman had her damaged left bronchus replaced by one constructed using her own stem cells. A section of the trachea from a donor was treated in order to remove the cells of the donor, leaving only a framework of collagen.

Adult stem cells derived from the woman's own bone marrow were then added to this collagen framework. Samples of these stem cells, together with cells derived from the lining of her trachea, were placed in a bioreactor for four days during which time they multiplied to coat the collagen framework. This was used to replace the damaged section of her bronchus. A month later the transplanted tissue had developed its own blood supply. This was claimed to be the first successful transplant using tissues derived from the patient's own stem cells.

(a) Outline the advantages of using a patient's own stem cells to treat a damaged bronchus rather than transplanting a complete bronchus.

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(b) Once most stem cells differentiate, they lose their ability to turn into other types of cells. However, some fully differentiated cells can be stimulated to change back into stem cells in tissue culture. Such cells are called induced pluripotent stem cells (iPS cells).

In an experiment with mice to produce iPS cells, it was discovered that the introduction of four genes would cause certain fully differentiated cells to change to iPS cells. These four genes indirectly result in the reprogramming of the differentiated cells by influencing other genes.

(i) Suggest **two** possible means by which the four genes were introduced into the cells.

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(ii) Outline how the addition of the four genes may influence the activity of many other genes to result in the reprogramming of cells.

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(iii) One of the steps in the process of developing induced pluripotent stem cells is the need to determine whether the cell line is pluripotent. One assay that has been used to determine the state of pluripotency is a Teratoma Forming Assay.

Teratoma are tumors of multiple lineages containing tissue derived from all three germ layers, which is unlike other tumors. Tumors are typically made up of only one cell type.

Suggest why the formation of a teratoma is indicative of a pluripotent state.

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(c) There is evidence to suggest that the introduction of the four genes mentioned in (b) caused an increase in the production of telomerase reverse transcriptase (TERT) in the fully differentiated cells.

Explain how TERT may help to change the cells back into stem cells.

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(d) A cybrid (cytoplasmic hybrid cell) is produced as shown in Fig. 1.1.

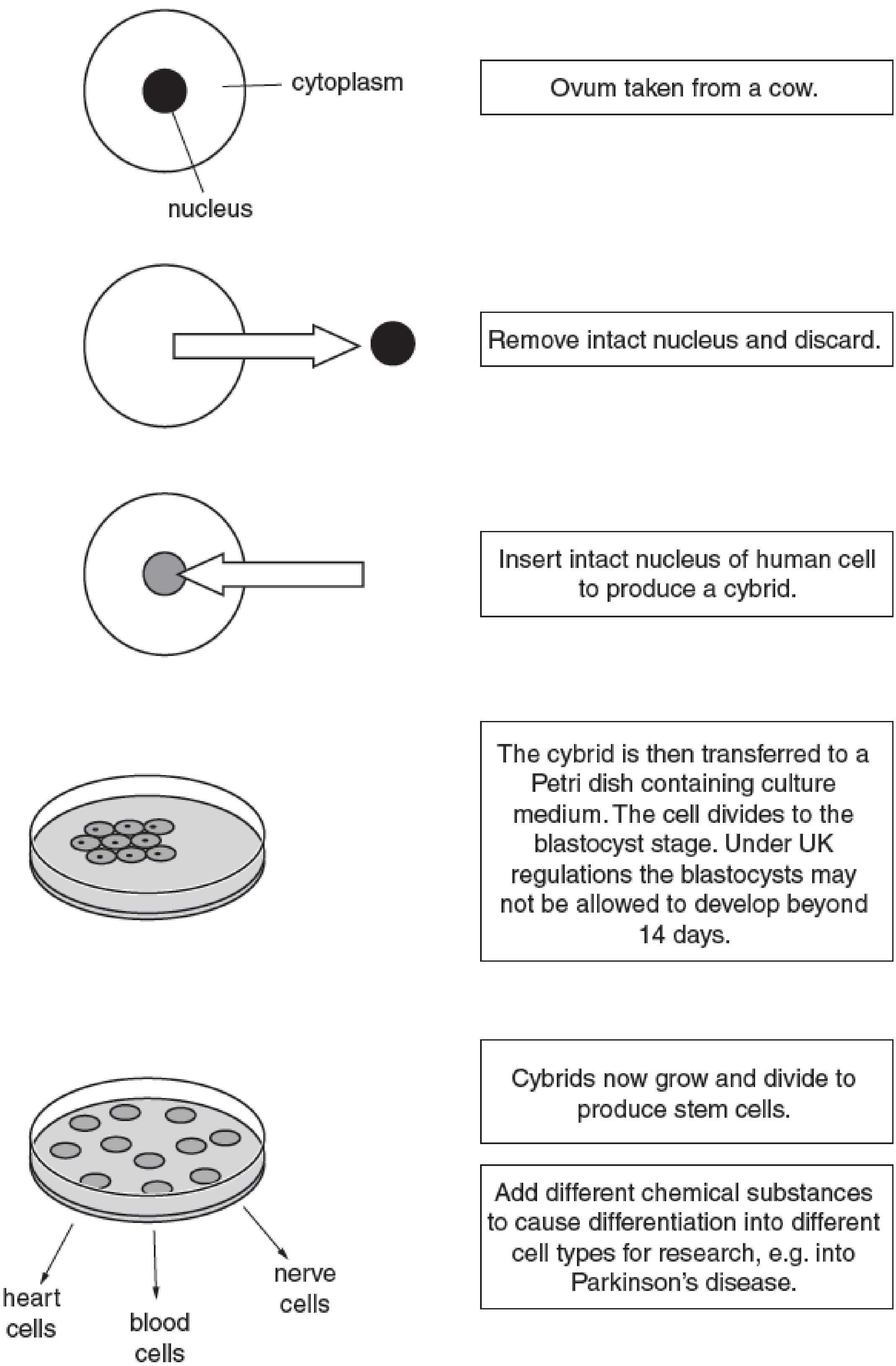

Fig. 1.1

(i) The DNA of such a cybrid is 99.6% human. The remaining 0.4% of the DNA is found in the cytoplasm. Explain why there is DNA in the cytoplasm of the cybrid.

…… [1]

(ii) When the Human Fertilisation and Embryology Bill was considered by the UK parliament in 2008, some people argued that it is unethical to allow the production of cybrids.

State whether you agree or disagree that this is unethical and explain why you reached this decision.

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[Total: 15 marks]

2 Golden Rice^{TM} is a genetically modified form of rice that produces relatively large amounts of β carotene in the endosperm. β carotene is metabolised in the human body to produce Vitamin A.

Fig 2.1 shows the metabolic pathway by which β carotene is synthesised in plants, and the enzymes that catalyse each step of the pathway.

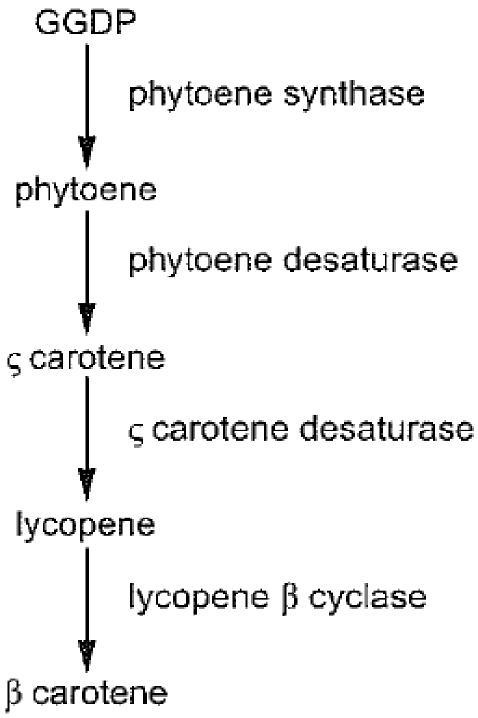

Fig. 2.1

The first types of Golden Rice^{TM} contained a phytoene synthase gene, *psy,* from daffodils and a gene *crtl,* which produced the two desaturase enzymes, from the bacterium *Erwinia uredovora.*

It was found that the first types of Golden Rice^{TM} produced only a very low mass of β carotene per gram of rice. Research continued to try to increase this.

Investigations were carried out to see if *psy* genes taken from species other than daffodils would enable rice endosperm to produce greater quantities of β carotene than the first types of Golden Rice^{TM}.

- Psy genes were isolated from the DNA of daffodil (Np), popcorn (Zm), chilli pepper (Ca), or tomato (Le). The genes were inserted into different plasmids.
- The promoter *Ubi1,* and *crt/* genes from *E. uredovora,* were also inserted into all of the plasmids.
- The four types of genetically modified plasmids were then inserted into different cultures of rice cells.
- The quantity of β carotene produced by these rice cells was measured.

Fig. 2.2 is a histogram showing the carotenoid content of maize calluses transformed with *psy* genes from different sources. C is the control plate without *psy* gene inserted into the maize calluses.

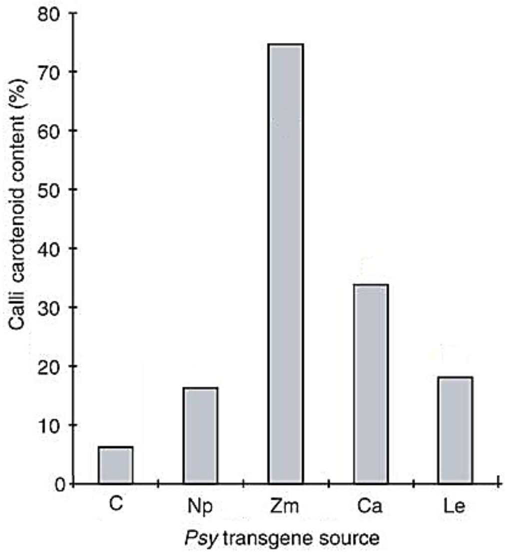

Fig 2.2

(a) Using Fig. 2.1 and 2.2, suggest which step of the metabolic pathway was limiting the production of β carotene.

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(b) With reference to Fig 2.2,

(i) evaluate the effectiveness of the various *psy* genes in increasing the carotenoid content of the maize calluses.

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(ii) suggest an explanation for the differences in the carotenoid content of the maize calluses.

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(c) Describe how a maize callus transformed with the plasmid containing the popcorn *psy* gene is formed.

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(d) Golden Rice™ are usually grown in isolated green houses away from conventional crops. Suggest reasons why this is so.

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[Total: 13 marks]

3 Restriction fragment length polymorphism, RFLP, refers to differences in fragment length generated by restriction enzymes. These differences are used to detect variations in genomes. This technique is widely used in molecular biology.

Fig 3.1 below shows an autosomal dominant disease segregating in a family. Genomic DNA from each family member was assayed for RFLPs for *Eco*RI sites, on Southern blots using probes derived from the gene.

The results for all individuals, except those for the foetus, are shown below in the pedigree chart. The sizes of the DNA fragments obtained are as indicated.

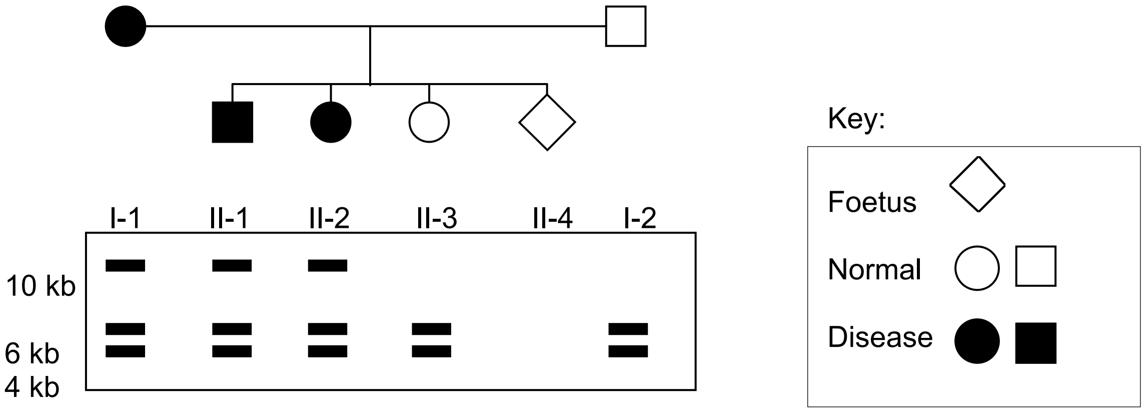

Fig 3.1

(a) **(i)** Sketch a suitable diagram to illustrate the region of genomic DNA recognised by the probe in individual **I-1**. Show the positions of *Eco*RI sites and the distance between them in kb.

[1]

(ii) Which RFLP fragment did the mother (**I-1**) contribute to the diseased children?

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(iii) What is the probability of the foetus having the disease? Explain your answer.

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Four individuals, I to IV, were tested to see if they are related.

The possible restriction sites of a gene and the results of four individuals after subjecting their DNA to restriction enzyme digestion are shown in Fig. 3.2.

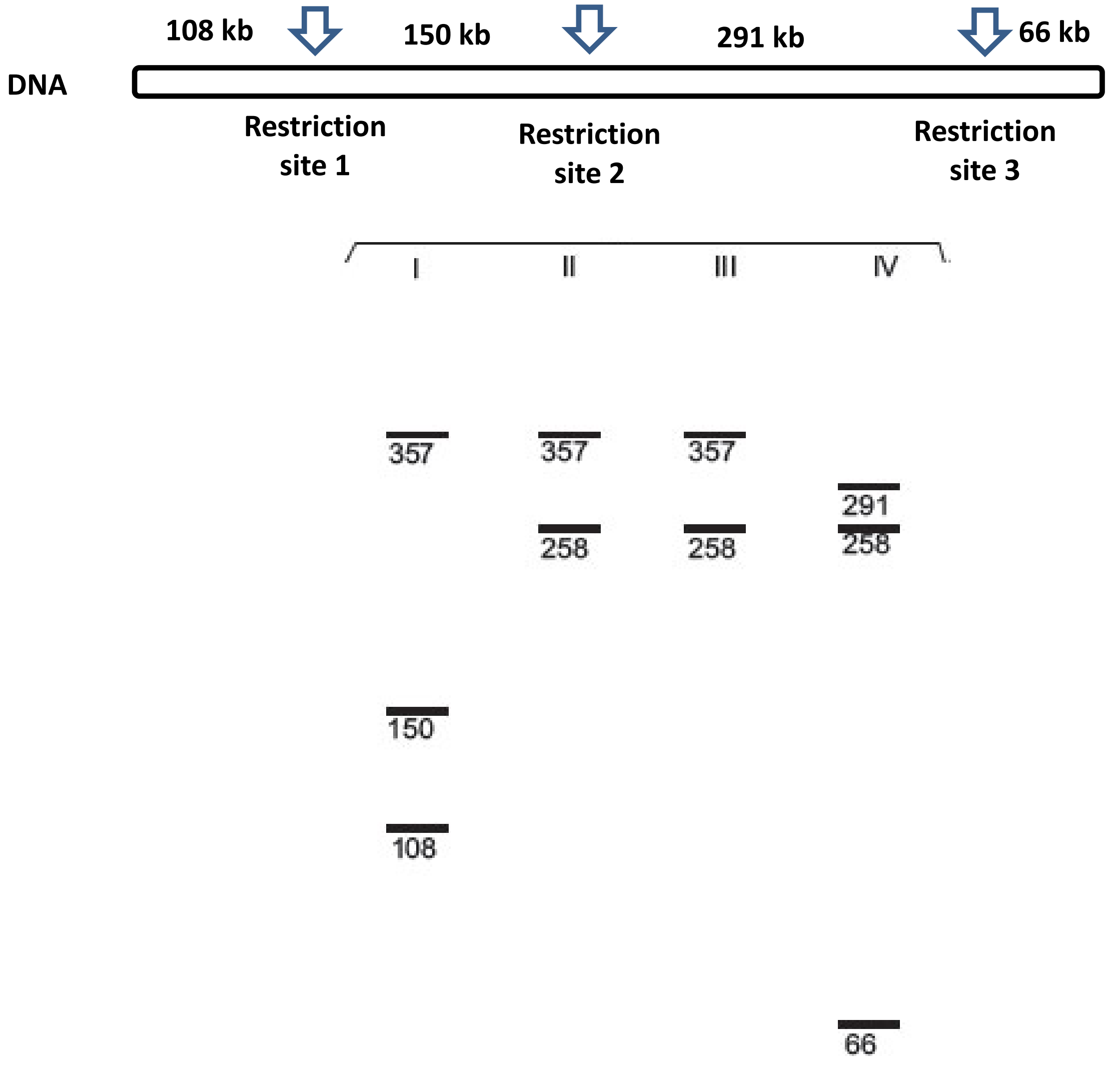

Fig. 3.2

(b) With reference to Fig. 3.2,

(i) state how many alleles does this gene exists in these four individuals. Explain your answer.

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(ii) Suggest why this gene can exist in more than one allelic form.

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(iii) Explain why individuals II and III cannot be parents of individual I.

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[Total:12 marks]

4 Planning question

P1 is a temperate phage that infects *Escherichia coli*. It is an unusual phage in that during the lysogenic stage, it's DNA remain transcriptionally silent and exists as an independent circular DNA plasmid-like element rather than integrates into the host genome like other temperate bacteria phage. In its lytic phase, P1 can produce transducing phages which carry part of the bacterial genome. Induction of P1 from the lysogenic cycle to the lytic cycle is known to occur over a temperature range of 32°C to 52°C.

You are required to plan, but not carry out, an investigation to determine the optimum temperature at which induction of P1 phage from the lysogenic cycle into the lytic cycle occurs.

Induction from the lysogenic to lytic cycle can be determined by observing the gene expression of β-galactosidase. β-galactosidase is an enzyme made up of two peptides, *lac Z-α* and *lac Z-Ω*, neither of which is active by itself but when both are present together, they spontaneously reassemble into a functional enzyme.

In mutant *E.coli* with a deletion in the *lac* operon, β-galactosidase lacking the *lac Z-α* is produced. The mutation can be repaired by transduction with P1 transducing phages which have been previously been reproduced in *E.coli* that do not contain the deletion in *lac* operon. These transducing phages contain DNA encoding *lac Z-α*.

Upon transduction of the mutant, *lac* operon deficient *E.coli* with transducing phages containing DNA encoding *lac Z-α*, the amount of β-galactosidase produced in the bacterial cell can be measured using a colorless chemical known as X-gal. Hydrolysis of X-gal by β-galactosidase will result in the formation of a blue product. The color intensity of the product can then be assayed using a colorimeter which measures the absorbance of different wavelengths of light in a solution in absorbance units (A.U). A solution of higher color intensity will absorb more light energy at a specific wavelength.

Your planning must be based on the assumption that you have been provided with the following equipment and materials, which you must use:

- a glucose-deficient liquid culture of mutant *E.coli* previously infected with P1 transducing phages carrying DNA encoding for *lac Z-α.* The transduced mutant *E.coli* culture is kept at 30°C. P1 phages are known to exist as lysogens in mutant *E.coli* cells at 30°C.
- lactose solution
- thermostatically regulated waterbath
- colorimeter with accompanying cuvettes
- liquid X-gal
- stopwatch
- sterile test tubes
- a variety of syringes for measuring volumes

Your plan should:

- have an explanation of theory to support your practical procedure
- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it
- include layout of results tables and graphs with clear headings and labels
- include full details and explanations of the procedures that you would adopt to ensure that the results obtained were as quantitative, precise and reliable as possible
- include relevant risks and precautions taken.

[Total:12 marks]

Free-response question

Write your answers to this question on the separate answer paper provided.

Your answer:

- should be illustrated by large, clearly labelled diagrams, where appropriate;
- must be in continuous prose, where appropriate;
- must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

5 **(a)** Describe how RFLP can determine completely linked genes and how RFLP analysis can be used in genome mapping. [6]

(b) *"Gene therapy clinical trials have mostly shown to be unsuccessful in terms of therapeutic interest."* Explain why. [7]

(c) Discuss the social and ethical concerns pertaining to the use of genetically modified animals. [7]

[Total: 20]